Standard Project Outline

1 Getting Started

In the standard project, we will learn how to apply causal models to real world data using statistical software. Specifically, we will be studying whether or not retirement benefits (specifically, the Canadian Old Age Supplement and pension (the OAS)) cause a decline in labour force participation rates. We will be doing this by working with the 2021 Canadian Census. Here are a few key terms:

- The **OAS** is a supplemental pension benefit available to Canadians to help them retire. It is only available to Canadian citizens or legal residents over 65 who have lived in Canada for at least 10 years.
- **Retirement**, in the context of the labour market, means moving from being in the labour force (employed or unemployed) to no longer being part of the labour force.

A simple model of the labour-leisure trade-off from intermediate or introductory microeconomics would predict that when a Canadian citizen or legal resident who has lived in Canada for at least 10 years, their labour force participation rate (LFPR) will suddenly, and discontinuously, change. The reason is the additional OAS benefits: the returns to retirement increase based solely on the individuals age. We would expect that benefits which encourage retirement would lead to more individuals choosing to retire. However, is this true?

1. A good place to start would be to *formalize* the labour-leisure model explained above. What confounding factors might play a role in whether a person chooses to retire or not, based on the model?

2. Once you feel like you have a good background in the basic model, think about an alternative. Why might we *not* see a decrease in LFPR after 65? What alternative models could explain such a finding, and what do they suggest might be important?

You should also keep in mind the details of the OAS, which we will discuss more in the readings in Section 2. In particular, keep in mind that non-immigrant Canadians, are always eligible for the OAS, while recent immigrants are not eligible for the OAS. This can be a useful feature for certain kinds of analyses.

2 Background

To get some background on the Canadian retirement system, you should read and review Ambrozas, Macdonald, and Masotti (2020) which can be found online here. A brief summary is that in Canada there are three "pillars" of retirement benefits.

1. Foundational Benefits, principally OAS.
2. Government Pensions.
3. Private Pensions and Savings.

We will focus mostly on (1) and (2) in our analysis. Detailed schedules can be found here. The first pillar is considered to be the "fundamental" one, since it is available to all eligible members regardless of work history or individual contributions to the public retirement system. OAS can be complicated, but the basics are:

- OAS is intended for medium-low income Canadians ages 65 and up. You are eligible at 65, and annual benefits increase yearly until age 75 if you defer.
- However, OAS is subject to a "recovery tax" once you earn more than about $90,000 a year in retirement, becoming zero once you earn more than $150,000. See more here.
- For very low income individuals or individuals above 75, OAS is supplement by either the Guaranteed Income Supplement (GIS) or a top-up amount, depending on your income.

This means that there is significant variation depending on both your age and income in terms of benefits from OAS and its supplemental programs.

The second pillar is work-related: the Canada or Quebec Pension plans. We will refer to both as the CPP, which is the more common of the two. The basic idea is that the CPP will replace about 33% of a contributor's annual income (capped at about $75,000).

- The CPP "vests" or becomes eligible for withdrawal starting at age 65. As with OAS, you can defer claiming it; a schedule is here.
- The amount of the CPP is based on your earnings over the lifespan of the program; i.e. until you retire. This means there is also considerable variation based on lifetime earnings.

Together, these two form the basis for much of the typical Canadian's retirement income. They are supplemented by the third pillar (private savings) which is quite heterogenous since it depends on personal attitudes, workplace benefits, and more.

At this point, you should try to place this setting in the potential outcomes model.

3 Data Set-Up

It is a good idea to do some preliminary clean-up to your dataset, to make it make your POM framework.

1. Download the data-file for this project; it uses the 2021 Census Data, which you should already have available.
2. You will want to construct an *analysis sample* which restricts the individuals in the Census so that we can focus on the research question. Think carefully about who you will want to “drop” from the analysis. Some suggestions or ideas include:
    - People who are older than 70 years old or younger than 60 years old.
    - People who have never worked or did not report a labour force status.
    - Non-permanent residents or individuals who did not report an immigration status.
    - Immigrants who did not report an age at immigration.
3. The outcome variable we are studying is labour force participation. You’ll need to create one; something like NILF (not in labour force).
    - Suggestion: think about how you should interpret your variable. Does it make sense?
4. We now need to form our two groups: the group which is eligible for treatment, and the ineligible group. Think carefully about who these should be. Suggestions might include:
    - An ineligible group will be immigrants who arrived age 60 years or older and are currently 60 years or older.
    - An eligible group will be immigrants who arrived prior to 60 years old, domestic Canadians, both of who are currently 60 years or older.
5. Now, think about your covariates. What will be important to collect? Do you have them already? Do you need to create them?

4 Checklist

Instructions: use this checklist to help organize your research project. Fill out your answers to the following questions.

1. What is your proposed research question?
2. Identify, both in terms of your project and the data, what the following variables are:
   - Outcome variable (Y_i)?
   - Treatment variable (D_i)?
   - Key covariates (X_i)
3. Why do you believe the covariates in (c) are important for inclusions?
4. Put your setting in the potential outcomes model.
5. What sample are you going to work with? Why?
6. Produce a simple table of descriptive statistics (mean, standard deviation, max, min) for the variables outlined in (2)
7. What method(s) are you going to use to answer your research question? We have (will) covered four in this course:
   - Hypothesis tests
   - Regression analysis
   - Difference-in-differences
   - Instrumental variables
8. For each of these methods, see some questions below to address the specifics.
9. How will your method (or methods) answer the question; be specific, in terms of a parameter or combination of parameters you will consider.
10. What do we need to believe in order for your answer to have a causal interpretation?
11. Write down 3-4 different specifications you intend to use to answer your question; write a short justification for each of them.
12. Run your specifications and produce a table of results. Interpret them in terms of answering your research question.
13. Brainstorm a possible extension or alternative analysis to your main analysis which would improve the causal interpretation of your results.

- 7a – Hypothesis Tests
  - Is it likely there is selection bias in your comparison? Explain it, and the direction
  - Can you identify what type (I, II, III) of selection bias is it?

  - How will you use your covariates?
  - What are the advantages and disadvantages of using a hypothesis test?

- 7b – Regressions
  - Why do you choose this specific specification?
  - What does the form of your specification say about the treatment effects?
  - Is there omitted variables bias in your analysis? To what extent do you covariates address or capture this?

- 7c – Difference-in-differences
  - What is the "time" dimensions? Group dimension? Why are these the right choices?
  - What is the common trends assumption in your setting? Is it credible?
  - How do you include covariates in your model? What are the limitations?

- 7d – Instrumental variables
  - What is your instrumental variable?
  - How does it meet the three assumptions of a good IV?
  - How should we interpret the treatment effect?

5 References

Ambrozas, Diana, Sonya Macdonald, and Sabrina Masotti. 2020. “Canada’s Retirement Income System.” 2019-40-E. Ottawa: Library of Parliament. www.ourcommons.ca/Content/LOP/ResearchPub 40-e.pdf.